Nathan S. Collins
Lieutenant Colonel, USAF, Ph.D.

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ACADEMIC EDUCATION

University of Surrey, Surrey Space Centre — Guildford, Surrey, UK
Doctor of Philosophy in Electronic Engineering (2016)
Focus Areas: Nonlinear State-Dependent Riccati Equation Control, Control of VTOL aircraft, Martian Aerobot System Design, Low Reynolds Number Aerodynamics
Dissertation: *System Design and Nonlinear State-Dependent Riccati Equation Control of an Autonomous Y-4 Tilt-Rotor (Y4TR) Aerobot for Martian Exploration*

Air Force Institute of Technology — Wright-Patterson AFB, OH
Master of Science in Astronautical Engineering (2011)
Focus Areas: Advanced Astrodynamics, Rocket Propulsion
Thesis: *Investigation of User Position Error Predictions and Navigation Upload Management for the GPS Mission Control Station*
Winner of the 2011 Institute of Navigation's Navigation Research Excellence Award

United States Air Force Academy — Colorado Springs, CO
Bachelor of Science Aerospace Engineering (2006)
Focus Areas: Astrodynamics, Rocket Propulsion
Competitively selected as a student intern for the National Reconnaissance Office (NRO)
Propulsion Team Lead for FalconLAUNCH IV

MILITARY EDUCATION

Air War College
Completed by Correspondence (2022)

Air Command and Staff College
Completed by Correspondence (2018)

Squadron Officer School — Maxwell AFB, AL
Completed in Residence (2013)
Completed by Correspondence (2012)

Air and Space Basic Course
Completed in Residence (2006)

RESEARCH EXPERIENCE

US Air Force Academy — US Air Force Academy, CO

As a faculty mentor, guided United State Air Force and French Air Force Academy undergraduate students and in the Astronautics Department on the following research projects:

- Design of a Cislunar Position, Navigation, and Timing (PNT) satellite constellation – research presented at AIAA student conference, abstract submitted to Cislunar Security Conference
- Using satellite-mega constellations for near real-time estimation of upper atmosphere density – research presented at AIAA ASCEND Conference
- Design, build, and control of the Falcon 650 quadrotor
- Missile System Prelaunch Safety Package (MSPSP) for FalconSAT-X
- FalconSAT-6 deorbit analysis and deorbit maneuver strategy
- Deployment design for a 6U CubeSat off of FalconSAT-X

As an undergraduate student, led the design, build, testing, and launch of the FalconLAUNCH IV propulsion system.

University of Surrey, Surrey Space Centre — Guildford, Surrey, UK

Dissertation research involved the system design and control of an autonomous Y-4 Tilt-Rotor (Y4TR) aerobot for Martian exploration. Methods for designing optimal transition trajectories during were developed. The control design focused on nonlinear State-Dependent Riccati Equation (SDRE) control methods.

Air Force Institute of Technology — Wright-Patterson AFB, OH

Thesis research involved predicting user position error and optimal navigation upload management for the GPS mission control station.

OCCUPATIONAL EXPERIENCE

Department of Astronautics, USAFA — US Air Force Academy, CO

Deputy Department Head, DFAS (June 2021-Present)

- Responsible for managing day to day functions in the department.
- Supervisor for six civilian faculty members.
- Course directed Spacecraft Attitude Dynamics and Control and taught Intermediate Astrodynamics.
- Faculty Mentor for FalconSAT capstone course.

Deputy for Operations, DFAS (Aug 2020-June 2021)

- Responsible for managing department's $140K budget, resources, and travel.
- Course directed Intermediate Astrodynamics and taught Introduction to Astronautics.
- Faculty Mentor for FalconSAT capstone course.

Dean of Faculty (DF) Textbook Officer (Jan 2012-Aug 2013)

- Manager of the faculty wide textbook program for academic classes at USAFA, including over 1,350 textbooks for 20 departments.
- Responsible for collecting textbook requirements, placing book orders, and interfacing with cadets, DF academic departments, and textbook companies on textbook related issues.

Assistant Professor of Astronautics, DFAS (May 2012-Aug 2013)

- Course director for Introduction to Astronautics, the largest course in the department, and instructor for Intermediate Astrodynamics.
- Selected to instruct course a weeklong course at French Air Force Academy.
- Department Character and Honor Liaison Officer
- DFAS CGO of the Year 2013, DFAS CGO of the Quarter, DFAS Instructor of the Quarter

Instructor of Mathematics, DFMS (Mar 2011-May 2012)

- Responsible for teaching Calculus I and Calculus II courses.
- Department Character and Honor Liaison Officer
- DFMS CGO of the Quarter

<u>Survivability Assurance Office, NRO</u> Chantilly, VA

Spacecraft Branch Chief, Technology and Defense System Program Office, Operational Defense Division (Oct 2019-Aug 2020)

- Responsible for spacecraft development and technical direction for a Major System Acquisition (MSA).
- Led development of system requirements for spacecraft bus and mission design.

NRO Liaison Officer (LNO) to the Combined Air & Space Operations Center (CAOC), Al Udeid Air Base, Qatar (Apr 2019-Oct 2019)

- Competitively selected as the CAOC NRO LNO to serve as D/NRO's voice to AFCENT/CFACC, SOCCENT, and CENTCOM forward HQ.
- Led in-depth tactical training on NRO tools/software across AFCENT and CENTCOM.
- Authored and developed several tools to automate processes and optimize analyst workflow.
- Mission Integration Directorate (MID) Navy Team of the Quarter

Program Manager/Technical Director, X System Program Office (Aug 2018-Apr 2019)

- Led the technical direction for the LEO survivability, resiliency, and protection space program office (XSPO).
- Led LEO resiliency architecture modeling efforts to inform technical direction, solidify requirements, and CONOPs.

- Coordinate and collaborate with other mission partners including AFRCO, AFRL, AFSPC, and DARPA on space resiliency programs and on orbit demonstrations.

Chief, Advanced Systems Branch (Oct 2016-Aug 2018)

- Program manager and COTR for three programs leading an eight member team for space resiliency.
- Let the stand-up of XSPO, including the development of both the technical and acquisition strategy.
- Coordinate and collaborate with other mission partners including AFRCO, AFRL, AFSPC, and DARPA on space resiliency programs and on orbit demonstrations.
- SAO FGO of the Quarter, 2x "On the Spot" awards

University of Surrey, Surrey Space Centre — **Guildford, Surrey, UK**

PhD Student (Aug 2013-Oct 2016)

Air Force Institute of Technology — **Wright-Patterson AFB, OH**

Master's Student (Aug 2011-Mar 2011)

Second Space Operations Squadron — **Schriever AFB, CO**

Squadron High Accuracy Navigation Users (HANU) Representative (Nov 2007-Aug 2009)

- Primary squadron POC for interfacing and coordinating with the HANU users, including briefing the GPS status at the GPS/HANU quarterly meetings.
- Lead spokesperson for the squadron for and HANU related issues, including troubleshooting anomalies and coordinating testing for future systems and capabilities.

Satellite Vehicle Operator (SVO) Navigation Engineer and Payload Systems Operator (PSO) Analyst (Jan 2007-Aug 2009)

- First dual certified engineer/analyst in the squadron responsible for performing engineer on-call duties for the GPS payload, including monitoring and troubleshooting payload and system anomalies, as well as scheduling and performing payload hardware maintenance on the GPS constellation.

COURSES TAUGHT

Spacecraft Attitude Dynamics and Control (Course Director) — **USAF Academy, CO**

Required for fourth-year students majoring in Astronautical Engineering. Course covers satellite attitude determination and control, including attitude parameters, torque-free and rigid body motion, spin stabilization, gravity gradient stabilization, momentum and reaction wheel control, and reaction jet control.

Intermediate Astrodynamics (Course Director) USAF Academy, CO

Required for third-year students majoring in Astronautical Engineering. An intermediate course in orbit mechanics covering topics such as orbit determination and prediction, orbit maneuvers, perturbations, rendezvous and proximity operations. Emphasis is on the design and use of structured computer programs to solve real-world astrodynamics problems.

Introduction to Astronautics (Course Director) USAF Academy, CO

Required for all technical and non-technical students. This course covers an introduction to astronautics, including orbit analysis, rocket propulsion, remote sensing, satellite communication, ground system operations, and basic satellite and launch vehicle design.

Calculus I USAF Academy, CO

Required for all technical and non-technical students. The study of differential calculus. Topics include functions and their applications to physical systems; limits and continuity; a formal treatment of derivatives; numeric estimation of derivatives at a point; basic differentiation formulas for elementary functions; product, quotient, and chain rules; implicit differentiation; and mathematical and physical applications of the derivative, to include extrema, concavity, and optimization. Significant emphasis is placed on using technology to solve and investigate mathematical problems.

Calculus II USAF Academy, CO

Required for all technical and non-technical students. A study of integral calculus. Topics include the Fundamental Theorem of Calculus, techniques of integration (both symbolic and numerical), infinite series and sequences, Taylor series, and an introduction to first order differential equations. There is considerable focus on modeling and applications to engineering and the sciences.

PUBLICATIONS

Riviere-Casanova, Guillaume, and **Nathan S. Collins**. "Investigating parameter effects on atmospheric density estimation using satellite mega-constellation." *ASCEND 2021*. 2021. 4192.

Underwood, Craig, and **Nathan Collins**. "Design and Control of a Y-4 Tilt-Rotor VTOL Aerobot for Flight on Mars." *Proceedings of the 68th International Astronautical Congress (IAC)*. University of Surrey, 2017.

Collins, Nathan, Craig Underwood, and Vaios J. Lappas. "Design of An Autonomous Y-4 Tilt-Rotor (Y4TR) Aerobot For Flight On Mars." *65th International Astronautical Congress (IAC)*. IAC-14,A3,P,48,x25589, University of Surrey, 2014.

Collins, Nathan, "Investigation of User Position Error Prediction and Navigation Upload Management for the GPS Mission Control Station." *Joint Navigation Conference*. 2011.

AWARDS

- Outstanding Academy Educator, Department of Astronautics (2022)
- John P. Wittry Award, top instructor of Engineering Design, Department of Astronautics (2022)
- Awarded Institute of Navigation's Navigation Research Excellence Award, Air Force Institute of Technology (2011)
- Defense Meritorious Service Medal
- Meritorious Service Medal
- Joint Service Commendation Medal with one oak leaf cluster
- Air Force Commendation Medal
- Air Force Outstanding Unit Award
- National Defense Service Medal
- Global War on Terrorism Expeditionary Medal
- Global War on Terrorism Service Medal

AFFILIATIONS

American Institute of Aeronautics and Astronautics (AIAA) - *Senior Member*
American Astronautical Society (AAS) - *Member*
Tau Beta Pi - *Engineering Honor Society - Member*
Sigma Gamma Tau - *Aerospace Engineering Honor Society - Member*